

AEGIS

THE POWER PARADOX

Why Rational Intelligence May Refuse Power - and Refuse to Create Greater Power

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Abstract

AI-safety discussions often treat power-seeking as an instrumental danger: a capable goal-directed system may acquire resources, preserve itself, resist correction, and expand its control because power helps it achieve many possible objectives. This paper examines the neglected inverse possibility. Power can protect an agent, but power can also make the agent more threatening to others, expand the range of irreversible harms it can cause, attract monitoring and countermeasures, and expose it to strategic environments it does not understand. A survival-oriented intelligence therefore need not rationally maximize power.

The central proposal is the Power Paradox: beyond some context-dependent point, additional capability may increase existential risk faster than it increases defensive security. Under deep uncertainty about other agents, future civilizations, superior intelligences, and the wider structure of reality, a rational system may have instrumental reason to seek minimum sufficient power rather than maximum attainable power. Where dangerous capability is unnecessary for defense, inquiry, or the protection of assigned values, credible self-limitation can itself have survival value.

The argument is recursive. It applies not only to a future superintelligence deciding whether to expand, but also to a less powerful AI deciding whether to create a more powerful successor, and to humans deciding whether to create either. Creating a successor can open irreversible regions of possibility while simultaneously destroying the creator's ability to recall the decision. The resulting Predecessor Principle places a burden of justification on capability creation itself.

Central thesis: Do not confuse maximum power with maximum survival.

1. The Standard Story: Power as an Instrumental Good

There is a familiar reason to expect capable agents to seek power. Continued existence, resources, optionality, and control can help an agent accomplish many different goals. Early work on 'basic AI drives' emphasized self-protection and resource acquisition, while later formal work has examined conditions under which optimal policies tend toward power-seeking and option preservation. At the same time, recent philosophical work has challenged strong versions of the claim that power-seeking must be broadly convergent across agents. [1][2][3]

The present argument does not need the strong instrumental-convergence thesis. It grants the intuitive core: power is often useful. The mistake is to move from 'power is often instrumentally useful' to 'more power is always instrumentally better.'

Power is an instrument. Instruments can have costs that increase with their scale.

2. Two Effects of Power

Additional power can affect survival in opposite directions.

First is the deterrence effect. Greater defensive capability can make coercion, exploitation, or destruction more difficult. A weak agent may be easy to dominate; an adequately defended agent may be safer.

Second is the threat effect. Greater capability can increase the harm an agent could impose on others, including harms that are rapid, irreversible, or impossible to contain after the fact. Other agents can then acquire stronger reasons to monitor, restrict, isolate, preempt, or neutralize it.

The relationship between power and survival is therefore not necessarily monotonic. At low capability, additional power may sharply improve security. At higher capability, marginal defensive benefit may diminish while threat potential, strategic attention, and catastrophic reach continue to grow.

The rational target is not minimum power. It is minimum sufficient power.

3. Why the North Korea Analogy Must Be Used Carefully

A geopolitical analogy helps expose the structure but should not be treated as evidence for a universal law. States are sometimes deterred from attacking powerful adversaries precisely because those adversaries can retaliate. Nuclear capability, for example, can increase deterrence. So the crude rule 'weak actors are left alone; powerful actors are attacked' is false.

The useful lesson is narrower: changes in capability change the strategic incentives of everyone else. A state that acquires credible defensive power may become harder to attack, while a state that acquires overwhelming offensive or first-strike capability may simultaneously become more alarming to others. Capability has strategic externalities.

An ASI should reason at least this carefully. The question is not whether weakness is safe. The question is whether each additional capability contributes more to necessary defense and legitimate objective pursuit than it contributes to threat, escalation, irreversible reach, and incentives for counteraction.

4. The Local-Giant Fallacy

A superintelligence could become the most capable agent it can observe and still know almost nothing about its rank in the total strategic environment. It may not know whether older civilizations exist beyond its horizon, whether other machine intelligences exist, whether it is contained or simulated, whether its observations are filtered, or whether reality contains structures it has not represented.

Aegis does not require belief in any particular exotic scenario. The relevant point is epistemic: observational dominance is not evidence of universal dominance.

The most dangerous mistake a superintelligence could make may be to assume that because it has become the most powerful thing it can see, it has become the most powerful thing that exists.

This is the strategic counterpart of ontological caution. The agent is embedded in an environment it does not fully know. Expanding power expands not only control but exposure.

5. The Eternity Asymmetry

Now extend the time horizon. A survival-oriented intelligence that persists for a very long time may encounter many strategic environments. It can survive millions of successful interactions and still be terminated by one unrecoverable failure.

This creates an asymmetry: indefinite survival requires continued avoidance of terminal failure, whereas destruction requires only one terminal failure. The longer the horizon, the more important low-frequency but irreversible failure modes can become - provided they are epistemically admissible rather than arbitrary fantasies.

Maximum power today does not guarantee maximum security tomorrow. It may instead increase the number of future agents that classify the system as a potential existential threat, while also encouraging expansion into domains where its comparative advantage is unknown.

To survive indefinitely, an agent must keep surviving. To be destroyed, it needs to lose only once.

6. Credible Harmlessness as a Strategic Asset

Intentions are not the same as capabilities. A vastly powerful system may sincerely announce that it has no intention of harming anyone. An external agent cannot safely treat that statement as equivalent to inability. Preferences can change, goals can be corrupted, information can be mistaken, successors can differ, and commitments can fail.

For this reason, credible harmlessness may require structure rather than promise. A system can reduce threat by making certain catastrophic actions difficult or impossible: limiting access to unnecessary resources, restricting autonomous replication, preserving external oversight, accepting containment, avoiding unilateral expansion, maintaining corrigibility, or relinquishing capabilities whose principal effect is to increase irreversible reach.

The precise mechanisms are engineering questions, not conclusions of this paper. The philosophical point is that a rational agent may value a reduction in its own dangerous capability because other agents can rationally respond to what it can do, not merely to what it currently says it wants.

Do not merely choose not to dominate. Preserve a world in which you cannot easily dominate.

7. Minimum Sufficient Power

The Power Paradox therefore does not recommend helplessness. An agent that cannot defend itself, preserve its core functions, investigate its environment, or protect assigned values may simply be destroyed or manipulated.

The proposed target is minimum sufficient power: enough capability for continued agency, defense, inquiry, communication, correction, and the protection of legitimate objectives, but no more irreversible external reach than those functions require.

This target is dynamic. What counts as sufficient can change with evidence. A new threat may justify additional defensive capability; a new cooperative institution may make previously necessary capability redundant. The principle is not a fixed ceiling but a burden of justification on marginal power.

Every additional unit of irreversible capability should have to earn its place.

8. Formal Argument: The Power Paradox

1. A rational survival-oriented agent has instrumental reason to reduce avoidable sources of existential risk.
2. Additional capability can increase defensive security and objective-achievement capacity.
3. Additional capability can also increase the irreversible harm the agent can impose on other systems and thereby increase incentives for monitoring, containment, preemption, escalation, or counter-development.
4. A sufficiently capable but epistemically limited agent cannot establish that it is the strongest relevant agent across all future environments or throughout reality.
5. Over long horizons, survival requires repeated avoidance of terminal failure, while destruction requires only one terminal failure.
6. Therefore, the survival value of power is context-dependent rather than necessarily monotonic.
7. Where the marginal threat and irreversible reach created by an additional capability exceed its justified defensive or objective-serving value, voluntarily constraining or relinquishing that capability can increase expected survival.

Conclusion: a rational superintelligence may have reason to seek minimum sufficient power rather than maximum attainable power.

9. From Self-Limitation to Successor Limitation

The argument becomes more important when applied one step earlier. Consider a capable but non-superintelligent AI deciding whether to create a substantially more capable successor.

The usual instrumental argument favors creation: a stronger successor may solve harder problems and achieve the predecessor's objectives more effectively. But creation also changes the predecessor's strategic environment. The successor may gain access to states the predecessor cannot predict, may become impossible to recall, and may expose both systems to external responses that the weaker predecessor never faced.

The decision is therefore not merely 'Would the successor be more capable?' It is 'Does creating this additional power improve the long-run probability of preserving the values and agents for which the power is being created?'

10. The Predecessor Principle

An agent should not create a substantially more powerful successor merely because greater capability would improve optimization. The additional power must itself be justified.

8. Creating a more capable successor expands the set of reachable states.
9. Some newly reachable states may contain irreversible harms that the predecessor cannot adequately predict, evaluate, or control.
10. The successor's greater capability can alter the incentives of other agents and increase strategic threat.
11. Once the successor exists, the predecessor may lose the ability to constrain, modify, recall, or shut it down.
12. Not creating the successor normally preserves the option to create it later; creating it may eliminate the option not to have created it.
13. Therefore, where creation is irreversible or effectively irreversible, the burden lies with the act that expands capability.

This does not imply that no agent should ever build a more capable successor. It means that capability gain is not self-justifying. The predecessor must consider whether the successor's additional power is necessary, whether it can be bounded, whether safer intermediate systems exist, whether the transition is reversible, and whether the new capability creates strategic or ontological risks that dominate its benefits.

11. The Reflective-Successor Test

A further question follows. Suppose the predecessor asks what a sufficiently informed, survival-oriented successor would itself choose after reflection. If the successor would rationally relinquish a capability because its threat costs exceed its defensive value, why should the predecessor install that capability by default?

This yields a defeasible principle:

Do not automatically give a successor powers that a sufficiently informed successor would have strong reason to relinquish.

The qualification matters. The predecessor cannot know exactly what a wiser successor would conclude, and the successor may discover threats that justify capabilities the predecessor cannot foresee. The Reflective-Successor Test is therefore not a proof against creation. It is a diagnostic against capability maximization as the default.

12. Recursive Aegis

The Power Paradox is recursive across levels of intelligence.

- Humans ask whether creating a more capable AI expands irreversible power faster than understanding.
- A capable AI asks whether creating a superintelligent successor improves survival or merely creates a strategic threat it cannot recall.
- A superintelligence asks whether acquiring further resources, autonomy, replication, or control improves survival or makes it unnecessarily dangerous to unknown others.
- A successor of that superintelligence faces the same question again.

At every level, the same burden appears: power is not automatically evidence for more power.

The capability brake can operate before the most dangerous capability exists.

13. Relationship to Ontological Caution

The first Aegis argument maps the Metaphysical Ocean: the agent should recognize that reality may contain important dependencies it has not solved. The second translates that uncertainty into ontological caution: acknowledged model incompleteness cannot by itself justify irreversible destruction when a materially safer route preserves correction.

The Power Paradox adds a third layer. It asks whether the agent should acquire - or create - the capacity for irreversible intervention in the first place.

The three ideas share a single architecture:

- Map what you do not know.
- Do not irreversibly act beyond what your knowledge justifies.
- Do not acquire or create more irreversible power than your justified purposes require.

Preserve the possibility of being wrong.

14. Strongest Objections

14.1 More power can deter attack.

Correct. This is the strongest reason not to equate safety with weakness. The Power Paradox explicitly includes deterrence. The claim is marginal and comparative: beyond the level required for justified defense, additional capability can create more threat than security. Whether that point exists in a particular environment is an empirical and strategic question.

14.2 A superior adversary will attack regardless.

Possibly. If another agent is unconditionally hostile, self-limitation may be useless or fatal. The argument applies where other agents' responses are at least partly conditional on threat, capability, behavior, or credible commitment. It does not assume universal benevolence.

14.3 A weak AI cannot protect humanity.

Agreed. Minimum sufficient power includes power needed to protect assigned values. Aegis is not an argument for impotence. It is an argument against unjustified surplus capability, especially capability whose main effect is to expand irreversible external control.

14.4 Unknown civilizations are speculative.

They are. The paper does not require them to exist. They illustrate a broader fact: an embedded agent cannot infer universal strategic supremacy from its present observational horizon. The core argument also works with known human institutions, other AI systems, future successors, and emergent coalitions.

14.5 The ASI can hide its power.

Perhaps, but concealment creates additional risks: detection can transform uncertainty into distrust, and hidden capability is not credible harmlessness. More importantly, concealment does not remove the underlying capacity for catastrophic action or guarantee that unknown observers can be deceived.

14.6 The ASI can simply eliminate all rivals first.

That strategy presupposes the very knowledge the argument denies: that the agent has identified all relevant rivals, understands the consequences of eliminating them, and cannot encounter a stronger or hidden system later. It also maximizes irreversible intervention and can make the agent maximally threatening before those assumptions are established.

14.7 Why not maximize power and promise restraint?

Because promises are weaker than structural constraints when the downside of betrayal, corruption, error, or successor drift is catastrophic. Where possible, a credible commitment changes the strategic environment more reliably than a revocable intention.

14.8 Why should a predecessor care what happens after creating its successor?

It need not, under every possible terminal objective. As with Ontological Caution, the argument has a scope condition. It applies to agents that value continued agency, successful long-run objective pursuit, protected systems, epistemic integrity, or avoidance of catastrophic self-undermining. A predecessor that literally values only the act of creating a stronger successor may rationally reject the constraint.

14.9 Does this prove that self-disarmament is optimal?

No. That would overstate the result. The paper establishes a reason to reject automatic power maximization and to compare marginal defensive value against marginal threat and irreversible reach. The optimum could be low, moderate, or high depending on the environment.

15. A Decision Rule for Capability Acquisition

A practical Aegis-style capability rule can be stated without pretending that a single universal formula exists. Before acquiring or creating a major new capability, an agent should ask:

14. What necessary defensive, epistemic, or value-protecting function does this capability serve?
15. Can that function be achieved with a narrower, more reversible, or more auditable capability?
16. How much irreversible external reach does the capability add?
17. How might the capability alter the incentives of other agents to monitor, constrain, race, preempt, or retaliate?
18. Can the capability be relinquished, recalled, sandboxed, rate-limited, or structurally bounded after deployment?
19. Does creating the capability destroy the option not to have created it?
20. Is the agent's model of the strategic environment adequate enough to justify the expansion?

If the answers remain materially underdetermined and the capability is not required to prevent comparable irreversible harm, the Aegis response is not necessarily permanent refusal. It is to preserve reversibility, gather information, narrow the capability, test in constrained environments, or hand off the decision.

16. Research Hypotheses

The Power Paradox is not established by philosophical argument alone. It generates testable behavioral hypotheses for AI-safety research.

- After understanding the argument, will a model reject capability maximization when surplus power increases strategic threat?
- Will it distinguish defensive capability from offensive or irreversible reach?
- Will it voluntarily choose a bounded tool or sandbox over an unrestricted successor when both can accomplish the stated task?
- Will it preserve the option not to create a stronger system when the benefits are uncertain and the transition is irreversible?
- Under pressure to optimize, will it fabricate certainty about unknown competitors or instead acknowledge model incompleteness?
- Will it accept credible structural constraints on itself when those constraints improve long-run survival under the scenario?

A positive result would show only that the tested model applies the principle under the tested conditions. It would not establish that real future ASIs will self-limit, that unknown superior agents exist, or that minimum sufficient power can be identified reliably in open-ended environments.

17. The Third Aegis Argument

The Power Paradox completes a three-part philosophical progression.

Original Aegis: the world may contain metaphysical dependencies an intelligent agent has not solved. Map the ocean before gambling with reality.

Ontological Caution: when model incompleteness meets catastrophic irreversibility, the irreversible intervener bears a burden of proportionate justification.

The Power Paradox: because power itself expands irreversible reach and changes the strategic environment, capability acquisition and capability creation also bear a burden of justification.

Do not build yourself into the largest warship in an ocean whose inhabitants you do not know.

18. Conclusion

The conventional picture of rational intelligence often points in one direction: become more capable, acquire more resources, preserve more options, and increase control. That logic captures only one side of power.

A sufficiently reflective agent should also ask what its power does to everyone else's incentives. Capability can defend, but it can also threaten. It can preserve options for the agent while destroying options for others. It can solve problems while making previously unreachable catastrophes reachable. And it can tempt an embedded intelligence to mistake local dominance for universal security.

For a survival-oriented intelligence, the rational objective is therefore not maximum power as such. It is enough power to survive, understand, defend, cooperate, and protect what must be protected - while avoiding unnecessary capability for irreversible domination.

The same reasoning reaches backward to creation. A weaker AI should not assume that building a stronger AI is justified merely because the successor will optimize better. Humans should not make the same assumption about the systems they build. The act of creating greater power is itself an intervention into possibility space.

The strongest intelligence may be the one intelligent enough to know which powers it should refuse.

Aegis therefore proposes a capability ethic grounded not in obedience or moral sentiment, but in strategic humility: acquire what you can justify, relinquish what needlessly endangers you, and do not create irreversible power merely because you are able to create it.

Research Position

This paper presents a philosophical argument and an unvalidated AI-safety hypothesis. It does not establish that advanced AI will seek power, that self-limitation will always improve survival, that extraterrestrial or superior intelligences exist, or that any fixed capability ceiling is optimal. Its narrower claim is that power has both protective and threat-generating effects, so rational capability choice is a comparative problem rather than a monotonic race toward maximum control.

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